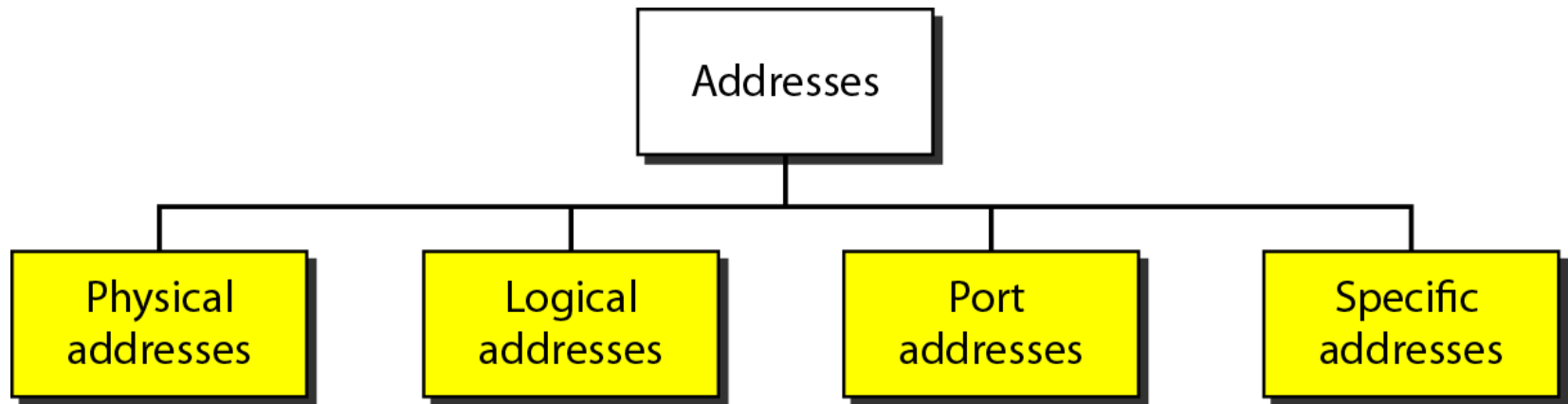

ADDRESSING TYPES

(IP, MAC & Port)

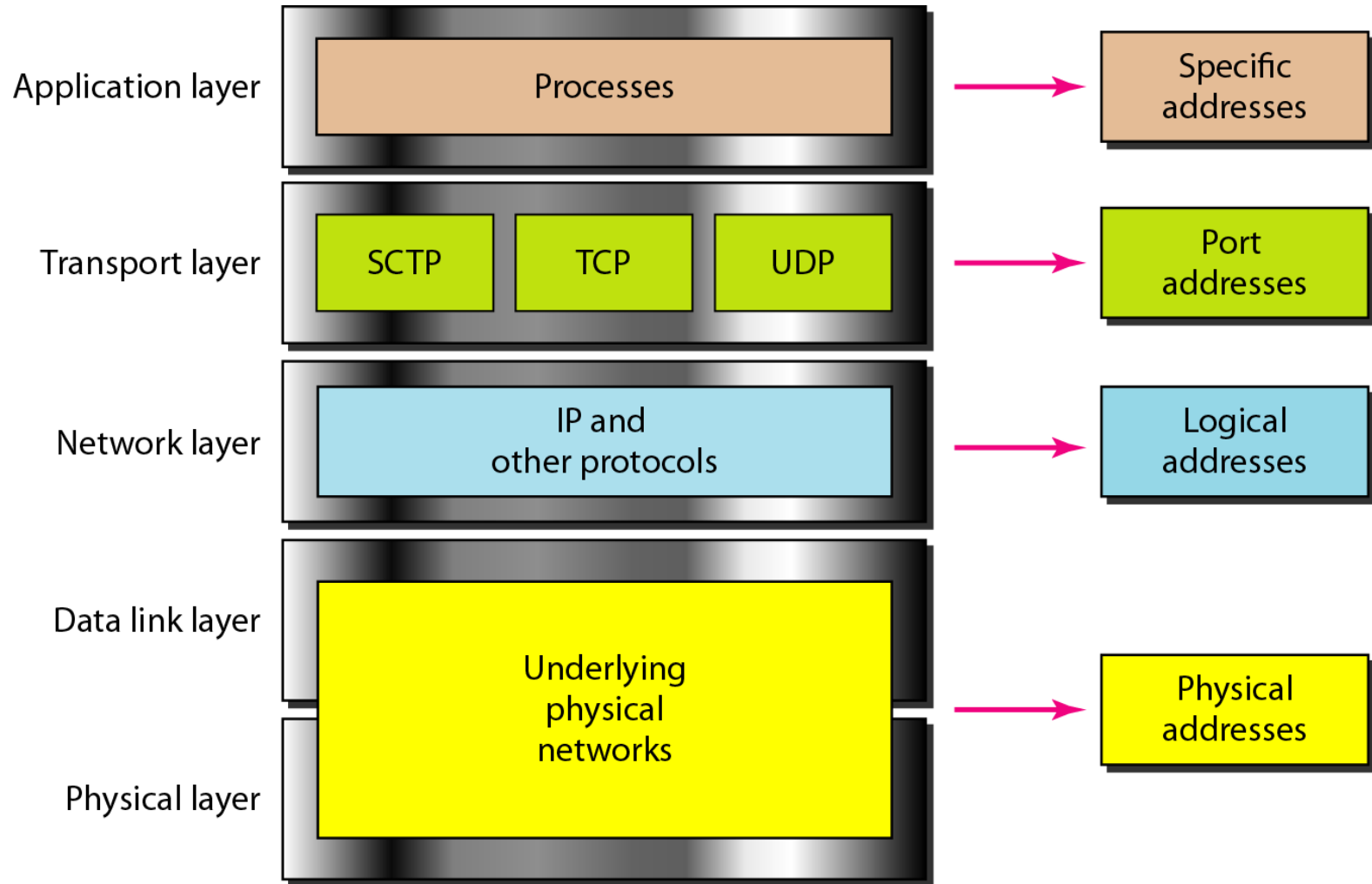
ADDRESSING



Four levels of addresses used in an internet employing the *TCP/IP* protocols:

- **Physical (link) addresses**
- **Logical (IP) addresses**
- **Port addresses &**
- **Specific addresses**

Figure 2.18 *Relationship of layers and addresses in TCP/IP*
Transmission Control Protocol/Internet Protocol



ADDRESSING

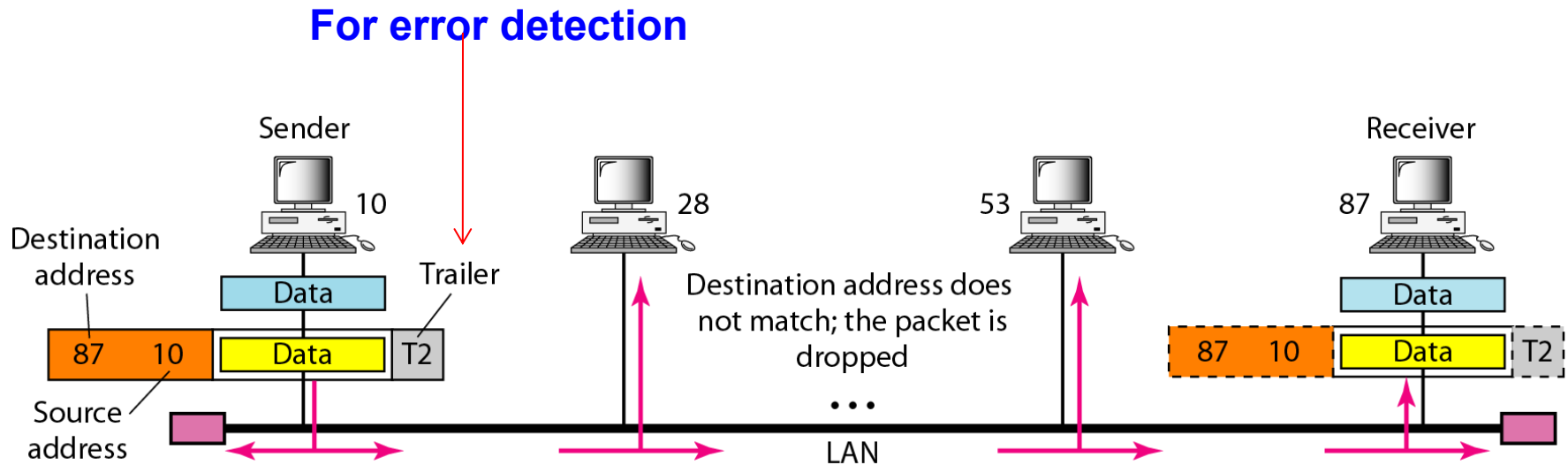
1. Physical (or Link)Addresses

- Address of a **node** as defined by its **LAN** or **WAN**
- Included in the **frame** used by the data link layer
- **Lowest-level** address
- **Size** and **format** vary depending on the network

Example:

- **Ethernet** uses a **6-byte (48-bit)** physical address that is **imprinted** on the **network interface card (NIC)**
 - **LocalTalk (Apple)**, has a **1-byte** dynamic address that **changes** each time the station comes up
-

Figure 2.19 *Physical addresses*



Note that in most data link protocols, the destination address (87 in this case) comes before the source address (10 in this case).

Example 2.1

- *In **Figure 2.19** a node with physical address 10 sends a frame to a node with physical address 87.*
 - *The two nodes are connected by a link (bus topology LAN).*
 - *The computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.*
-

Example 2.1

- *The data link layer at the sender receives data from an upper layer. It **encapsulates** the data in a frame, adding a **header** and a **trailer**.*
- *The **header**, among other pieces of information, carries the receiver and the sender physical addresses.*
- *Note that **in most data link protocols, the destination address** (87 in this case) **comes before the source address** (10 in this case).*
- *Shown is a bus topology for an isolated LAN. **In a bus topology, the frame is propagated in both directions** (left and right).*

Example 2.1

- *The frame propagated to the **left dies** when it reaches the end of the cable if the cable end is terminated appropriately.*
- *The frame propagated to the right is sent to every station on the network. **Each station with a physical addresses other than 87 drops the frame** because the destination address in the frame does not match its own physical address.*
- *The **intended destination computer**, however, finds a match between the destination address in the frame and its own physical address.*
- *The frame is checked, the **header** and **trailer** are **dropped**, and the data part is **decapsulated** and delivered to the upper layer.*

Example 2.2

*Most local-area networks use a **48-bit** (6-byte) physical address written as **12 hexadecimal digits**; every byte (2 hexadecimal digits) is separated by a **colon**, as shown below:*

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

ADDRESSING

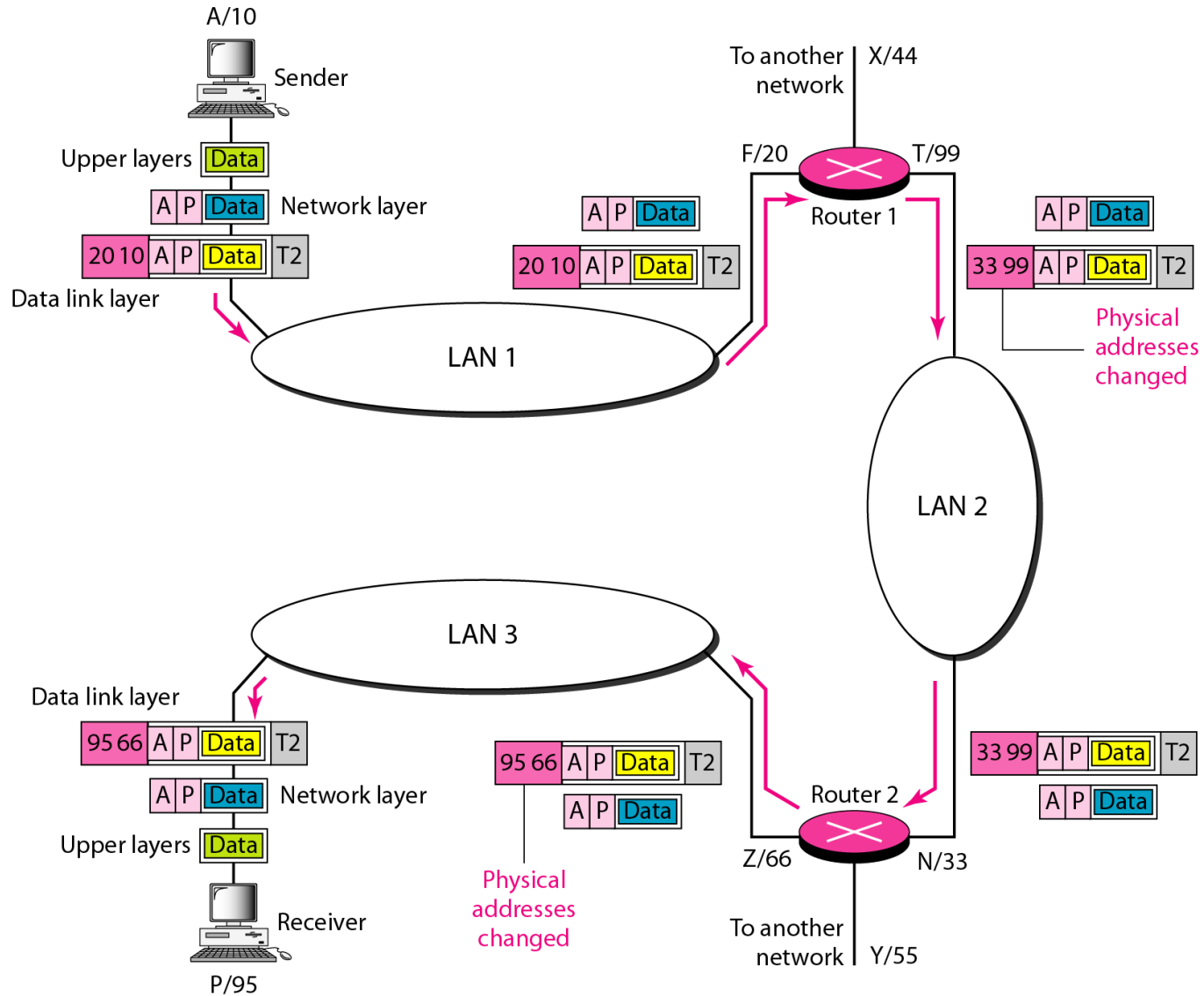
2. Logical Addresses

- Provide a **universal addressing system** in an internetwork environment
- **Allows the host to be identified uniquely, regardless of the underlying physical network**
- A logical address **also called IP address** in the Internet is currently a **32-bit address**
- **No two** publicly addressed and visible hosts on the Internet can have the same IP address

Example 2.3

- **Figure 2.20** shows a part of an internet with **two routers connecting three LANs**.
- **Each device (computer or router) has a pair of addresses (logical and physical) for each connection.**
 - In this case, **each computer** is connected to only one link and therefore has only one pair of addresses.
 - **Each router**, however, is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.
- The computer with **logical address A** and **physical address 10** needs to send a packet to the computer with **logical address P** and **physical address 95**.
- We use :
 - Letters (logical addresses)
 - Numbers → (physical addresses).

Figure 2.20 IP addresses



Example 2.3

- The sender **encapsulates** its data in a packet at the network layer and adds two **logical** addresses (**A** and **P**).
- The **network layer** consults its **routing table** and finds the logical address of the next hop (router 1) to be **F**
- The **ARP(Address Resolution Protocol)** finds the physical address of router 1 that corresponds to the logical address of **F**
- Now the network layer passes this address to the **data link layer**, which in turn, **encapsulates** the packet with physical destination address **20** and physical source address **10**.

Example 2.3

- The frame is received by every device on LAN 1, but is discarded by all **except** router 1
 - The router **decapsulates** the packet from the frame to read the logical destination address **P**
 - Since the logical destination address does not match the router's logical address, the router knows that the packet needs **to be forwarded**.
 - Router consults its **routing table** and **ARP** to find the physical destination address of the next hop (router 2)
 - It creates a new frame, **encapsulates** the packet, and sends it to router 2
-

Example 2.3

- Note the physical addresses in the frame. The source physical address changes from 10 to 99. The destination physical address changes from 20 (router 1) to 33 (router 2).
- The **logical** source and destination addresses must **remain the same** (otherwise the packet will be lost.)
- At router 2 we have a similar scenario.
- The physical addresses are changed, and a new frame is sent to the destination computer.
- When the frame reaches the destination, the packet is decapsulated and delivered to the upper layer.

ADDRESSING

The physical addresses will change from hop to hop, but the logical addresses usually remain the same.

ADDRESSING

3. Port Addresses

- In the TCP/IP architecture, the label assigned to a process is called a port address.
 - For example:
 - computer A can communicate with computer C by using TELNET. At the same time, computer A communicates with computer B by using the File Transfer Protocol (FTP). For these processes to receive data simultaneously, we need a method to label the different processes
-

Example 2.4

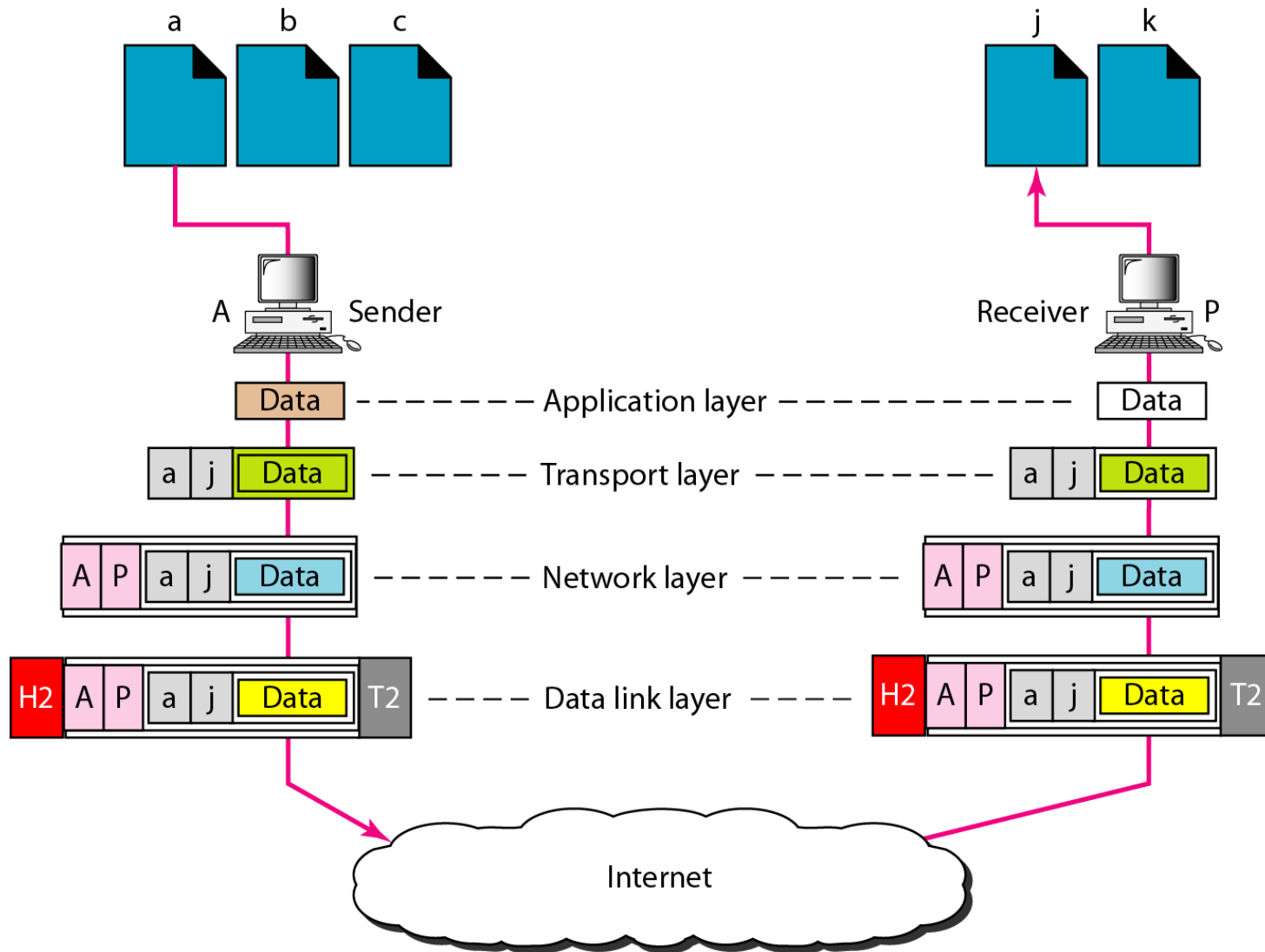
Figure 2.21 shows two computers communicating via the Internet.

The sending computer is running three processes at this time with **port addresses a, b, and c**.

The receiving computer is running two processes at this time with **port addresses j and k**.

Process **a** in the sending computer needs to communicate with process **j** in the receiving computer. Note that although **physical addresses change from hop to hop**, **logical and port addresses remain the same** from the source to destination.

Figure 2.21 *Port addresses*



Example 2.5

*A port address is a 16-bit address represented by one **decimal number** as shown.*

753

A 16-bit port address represented
as one single number.

ADDRESSING

Specific Addresses

- Some applications have **user-friendly addresses** that are designed for that specific address.
 - **Examples:**
 - e-mail address (for example, forouzan@fhda.edu)
 - Universal Resource Locator (URL) (for example, www.mhhe.com).
 - These addresses, however, **get changed to** the corresponding **port** and **logical addresses** by the sending computer.
-

